Benchmark results

NewTOS on Raspberry PI 4

unknown AArch64 TOS 4.04 1920x1080 256c blit none

115 of 115 tests run, 1.0 s each. Values are operations per second (higher is better). Bars compare tests within each group.

Groups

VDI primitives	38 tests run
AES windows & events	21 tests run
Alignment (blitter)	20 tests run
Full-screen windows	17 tests run
GDOS fonts	14 tests run

VDI primitives

Test	ops/s	Note
Line, 2 points	23516	
Polyline, 32 points	791.4	
Wide line (7px, round)	4034.4	
Dashed line	21152	
User line style	22649	
XOR line	20572	
Markers x16	23743	
Filled rect 32x32	91871	
Filled rect, full window	578.2	
Bar, hatched + border	53843	
Pattern fill rect	90601	
User pattern fill	91444	
XOR filled rect	119300	
Polygon, 8 points	17553	
Filled circle	2496.5	
Filled ellipse	2870.0	
Arc	21251	
Pie slice	2559.0	
Elliptical arc	22649	
Elliptical pie	2924.7	
Rounded box	44830	
Filled rounded box	9989.2	
Contour fill 64x40	7278.6	
Text, 15 chars	30912	
Text, 84 chars	5388.5	
Text with effects	19411	
Text rotated 90	31206	
Justified text	19139	
Text extent query	399984	
Blit scr->scr 64x64	38823	
Blit scroll window	240.5	
Blit mem->scr 64x64	27971	
Blit scr->mem 64x64	24765	
Mono blit (vrt_cpyfm)	45055	
Transform form 64x64	n/a	not supported here
Get pixel	506557	
Hide/show mouse	111346	
Attribute calls x3	172462	

Open/close workstation	47034	

AES windows & events

Test	ops/s	Note
Window create+delete	146721	
Window open+close	227.6	
Window move	713.4	
Window resize	909.0	
Window top (2 windows)	460.5	
Set window title	1576.3	
Set info line	1583.8	
Set slider pos+size	2425.5	
Scroll window content	218.6	
wind_get work area	321253	
Rectangle list walk	145995	
wind_calc	303406	
wind_update pair	167649	
objc_draw dialog	4150.4	
objc_find	282211	
form_dial redraw 64x48	11206	
Menu bar redraw	5935.2	
graf_mkstate	322836	
graf_mouse off+on	87063	
Event poll (evnt_multi)	141105	
Message round trip	59361	

Alignment (blitter)

Test	ops/s	Note
Fill 64x64, aligned	115923	
Fill 64x64, x+3	46368	
Blit 64x64, aligned	57343	
Blit 64x64, src+3 dst+11	35445	
Mem->scr 64x64, aligned	28108	
Mem->scr 64x64, x+3	27701	
Mono blit, aligned	47260	
Mono blit, x+3	44830	
Text, x on 8 px	35445	
Text, x+3	29824	
Scroll up, aligned edges	221.8	
Scroll up, edges x+5	209.3	
Scroll left 16 px	186.3	
Scroll left 3 px	185.4	
Move window, aligned	635.8	work x mod 16 = 0
Move window, off-grid	620.3	work x mod 16 = 5
Open/close win, aligned	225.4	work x mod 16 = 0
Open/close win, off-grid	223.0	work x mod 16 = 5
Scroll window, aligned	776.6	work x mod 16 = 0
Scroll window, off-grid	730.4	work x mod 16 = 5

Full-screen windows

Test	ops/s	Note
Open/close full window	134.2	work 1896x1018
Full <-> normal size	241.7	
Top full / test window	278.6	work 1896x1018
Fill full window	377.3	work 1896x1018
Scroll up 1 line	176.8	work 1896x1018
Scroll left 16 px	181.0	work 1896x1018
Scroll left 8 px	130.0	work 1896x1018
Scroll left 3 px	124.5	work 1896x1018
Scroll text 8pt	173.2	system font
Scroll text 10pt	176.8	system font
Scroll text 12pt	176.8	got 10pt, system font
Scroll text 14pt	176.8	got 10pt, system font
Scroll text 18pt	176.0	system font
Scroll text 24pt	184.5	got 20pt, system font
Scroll text 36pt	184.5	got 20pt, system font
Scroll text 48pt	184.5	got 20pt, system font
Scroll, size per line	178.5	system font

GDOS fonts

Test	ops/s	Note
Text 6pt	47952	got 8pt, system font
Text 8pt	48187	system font
Text 10pt	23293	system font
Text 12pt	23183	got 10pt, system font
Text 14pt	23516	got 10pt, system font
Text 18pt	16061	system font
Text 24pt	8652.5	got 20pt, system font
Text 36pt	8693.3	got 20pt, system font
Text 48pt	8734.5	got 20pt, system font
Text 72pt	8572.0	got 20pt, system font
Text 10pt bold	22754	system font
vst_point, 8 sizes	399984	
Font switch + vst_point	n/a	not supported here
Arbitrary size + text	n/a	not supported here
Scroll, mixed sizes	220.7	system font
Text extent at 12pt	399984	
Enumerate font names	n/a	not supported here
Load+unload fonts (x3)	n/a	not supported here

Configuration

Machine

```
Description     : NewTOS on Raspberry PI 4
Machine type    : unknown
CPU             : AArch64 (native)
FPU             : none reported
Blitter         : not present
ST RAM          : 14336 KB total
TT/Fast RAM     : 729088 KB total
```

Operating system

```
TOS             : 4.04  (11/08/2026 m/d/y)
Country / video : USA, NTSC
GEMDOS          : 0.48
MiNT            : not running
Largest free    : ST 10240 KB
                  TT/Fast 715424 KB
```

AES / environment

```
AES version     : 3.40
Multitasking    : no (max apps 1)
Theme pens      : green 166, red 122, blue 149, cyan 155, highlight 106
Application id  : 0
appl_getinfo    : not available
Desktop area    : 0,19 1920x1061
AES char / box  : 8x16 / 19x19
Current dir     : C:\APPS\FBENCH
PATH            : /dev/c/
```

Display

```
Resolution      : 1920x1080, Getrez 4 (TT Medium)
Colours         : 256 on screen, 16 planes
Palette         : 0 (direct colour)
Lookup table    : no (true colour)
Pixel size      : 278 x 278 microns
Screen memory   : 4147200 bytes at $3dbf5000
VDI handles     : physical 1, ui 3 (bench ws opened only for runs)
VDI max vertices: 1024
Text rotation   : 90 degree steps
Writing modes   : 4
```

GDOS / fonts

```
GDOS            : none (system font only)
Fonts           : 1 (0 loaded by GDOS)
  > id     1  6x6 system font
```

Cookie jar

```
NTOS _MCH _IDT _AKP EdDI
```